

Madrasa-e-Riyaziyat 2026

Lesson Plan

1 Modules

This years madarsa has two main modules:

- **Graph Theory Module:** Core mathematical foundations and structured problem solving related to graph theory
- **Miscellaneous Topics Module:** Various mathematics, puzzles, simulations, and other fun concepts.

2 Daily Schedule (tentative)

Each day is divided into two main sessions:

Time	Session
9:00 – 11:00	Graph Theory
11:00 – 1:00	Break
1:00 – 3:00	Miscellaneous Topics

3 Graph Theory Module

This module builds foundational understanding of graph theory through problems and applications.

Topics

1. Logic and Proofs (Introduction)
2. Bridge of Konigsberg
3. Art Gallery Problem
4. 3 Utilities Problem
5. Map Coloring Problem
6. Graph Reconstruction Problem
7. Moore's Problem / Hall's Marriage Theorem / Handshaking Lemma

4 Miscellaneous Topics Module

This module explores mathematical ideas through puzzles, simulations, and interdisciplinary topics.

Topics List

- Tilings, Tessellations, Escher Patterns, Symmetry, Fibonacci Sequence
- Chaos Theory (Double Pendulum Simulation)
- Galton Board (Normal Distribution Intuition)
- Encoding Systems: Morse Code, Barcodes, UPC
- Polygon Triangulation Puzzle
- Monty Hall Problem Simulation
- Sherlock Hat Puzzle
- Minimum Surface Ice Box Problem
- Sliding Puzzle Graph Representation
- Rubik's Cube and Combinatorics
- Automata Theory (Introduction)
- Knights Tour Problem
- Counting Techniques
- Number Theory Puzzles
- Godel's Incompleteness Theorem (Introductory Idea)
- Knot Theory and Topology (Basic Intuition)

5 Notes

- Topics may be adjusted based on student engagement and pacing.
- Graph theory module topics are confirmed; however the miscellaneous topics module has tentative topics, from which people can pick out or add their required topics.
- Miscellaneous topics can include live demonstrations or coding simulations if the lecturer wants,
- TThere will be instructors and educators coming in from IBA, LUMS and other universities to deliver lectures.